# Midterm Rev 2

0.5 +
1.0 +
1.0 +
0.25 +
1.0 +
1.0 +
0.9 +
1.5
~ 6.25 + 1 = 7.25

1. $A \subseteq \mathbb{R}$ bounded set.

$\forall \varepsilon > 0, \exists x \in A$ s.t. $x < \inf(A) + \varepsilon$ ?

~0.5/1

$\inf(A) \le x, (\forall) x \in A$

$\left.\begin{array}{l} \text{if } y \le x, (\forall) x \in A \\ y \in lb(A) \end{array}\right\} \Rightarrow y \le \inf(A) \quad (1)$

Let $\varepsilon > 0 \Rightarrow \inf(A) + \varepsilon > \inf(A) \overset{(1)}{\Longrightarrow} \inf(A) + \varepsilon \notin lb(A)$

$\Rightarrow \exists x \in A$ s.t. $\inf(A + \varepsilon) > x$

$\Rightarrow \exists x \in A$ s.t. $x < \inf(A + \varepsilon)$

(I can't really get the logic of this :( )

2. a) $\sum_{n \ge 1} \frac{a^n}{a + n^2}, a > 0$ convergence?

$x_n = \frac{a^n}{a + n^2}$

1p

$\frac{x_{n+1}}{x_n} = \frac{\cancel{a^{n+1}}\, a}{a + n^2 + 2n + 1} \cdot \frac{a + n^2}{\cancel{a^n}\, 1} = \frac{a(n^2 + a)}{n^2 + 2n + a + 1} = a.$

If $a < 1 \Rightarrow$ conv.

$a > 1 \Rightarrow$ div.

$a = 1 \Rightarrow x_n = \frac{1}{n^2 + 1} \Rightarrow \sum_{n \ge 1} \frac{1}{n^2 + 1}$ "like" $\sum \frac{1}{n^2}, 2 > 1$ convergent.
$\Rightarrow$ conv.

$\lim_n \frac{\frac{1}{n^2+1}}{\frac{1}{n^2}} = \lim_n \frac{n^2}{n^2 + 1} = 1 \in (0, \infty) \Rightarrow$ same nature.

Conclusion $\to$ conv, $(\forall)\, a \le 1$
$\to$ div, $(\forall)\, a > 1$.

b) $x_n = \sum_{n \ge 1} \frac{2 \cdot 4 \cdot \ldots \cdot (2n)}{1 \cdot 3 \cdot \ldots \cdot (2n-1)}$

1p

$\frac{x_{n+1}}{x_n} = \frac{\cancel{2 \cdot 4 \cdot \ldots \cdot (2n)}\,(2n+2)}{\cancel{1 \cdot 3 \cdot \ldots \cdot (2n-1)}\,(2n+1)} \cdot \frac{\cancel{1 \cdot 3 \cdot \ldots \cdot (2n-1)}}{\cancel{2 \cdot 4 \cdot \ldots \cdot 2n}} = \frac{2n+2}{2n+1} = 1$ inconclusive.

Raabe - Duhamel:

$$\lim_{n} n\left(\frac{x_n}{x_{n+1}}-1\right)=\lim_{n} n\cdot\left(\frac{2n+1}{2n+2}-1\right)=\lim_{n} n\cdot\left(\frac{\cancel{2n}+1-\cancel{2n}-2}{2n+2}\right)=$$

$$=\lim_{n} n\cdot\left(\frac{-1}{2n+2}\right)=-\frac{1}{2}<1 \text{ divergent.}$$

c) $\sum_{n\geq 1} n\left(\sin\frac{1}{n}-\sin\frac{1}{n+1}\right)$

Method 1

$$\sin\frac{1}{n}-\sin\frac{1}{n+1}=2\sin\frac{\frac{1}{n}-\frac{1}{n+1}}{2}\cos\frac{\frac{1}{n}+\frac{1}{n+1}}{2}=$$

~0.25/1p.

$$=2\sin\frac{\cancel{n}+1-\cancel{n}}{2\cdot n(n+1)}\cos\frac{2n+1}{2n(n+1)}$$

$$=2\sin\frac{1}{2n(n+1)}\cos\frac{2n+1}{2n(n+1)}$$

$\to \cos 0=1.$

$$\lim_{x\to 0}\frac{\sin x}{x}=1$$

$\downarrow$

$$\lim_{n}\frac{\sin\frac{1}{2n(n+1)}}{\frac{1}{2n(n+1)}}=1$$

We can compute with $n^2$ instead and make use of it!

$$\Rightarrow \lim_{n} n^2\left(\sin\frac{1}{n}-\sin\frac{1}{n+1}\right)=\lim_{n} n^2\cdot\frac{2}{2n(n+1)}=1.$$

$$\Rightarrow 1=\lim_{n} n^2\left(\sin\frac{1}{n}-\sin\frac{1}{n+1}\right)$$

$$=\lim_{n} n\cdot n\left(\sin\frac{1}{n}-\sin\frac{1}{n+1}\right)$$

$$=\lim_{n}\frac{n\left(\sin\frac{1}{n}-\sin\frac{1}{n+1}\right)}{\frac{1}{n}}=1\in(0,\infty) \Rightarrow \text{same nature.}$$

$$\Rightarrow \sum_{n\geq 1} n\left(\sin\frac{1}{n}-\sin\frac{1}{n+1}\right) \text{ „like" } \sum_{n\geq 1}\frac{1}{n^{1}}\geq 1 \Rightarrow \text{divergent.}$$

Method 2 : Mean-Value Theorem

$$\sin\frac{1}{n}-\sin\frac{1}{n+1}=\left(\frac{1}{n}-\frac{1}{n+1}\right)\underbrace{\cos\frac{1}{c_n}}_{\to 1 \text{ as } n\to\infty},\quad c_n\in(n,n+1)$$

$$=\frac{1}{n(n+1)}\cos\frac{1}{c_n}$$

$$\Rightarrow n\left(\sin\frac{1}{n}-\sin\frac{1}{n+1}\right)=n\cdot\frac{1}{n(n+1)}\cos\frac{1}{c_n}\simeq\frac{1}{n+1} \Rightarrow \text{divergent series.}$$

Method 3 : $\displaystyle\sum_{n=1}^{N} n\left(\sin\frac{1}{n} - \sin\frac{1}{n+1}\right) = \sum_{n=1}^{N} n\sin\frac{1}{n} - \sum_{n=1}^{N} n\cdot\sin\frac{1}{n+1}$

$$= \sum_{n=1}^{N} n\sin\frac{1}{n} - \sum_{n=1}^{N}(n+1)\sin\frac{1}{n+1} + \sum_{n=1}^{N}\sin\frac{1}{n+1} = S_1 - S_2 + S_3.$$

$$= 1\cdot\sin\frac{1}{1} + \cancel{2\sin\frac{1}{2}} + \cancel{3\sin\frac{1}{3}} + \dots + \cancel{N\sin\frac{1}{N}}$$

$$- \cancel{2\sin\frac{1}{2}} - \cancel{3\sin\frac{1}{3}} - \dots - \cancel{N\sin\frac{1}{N}} - (N+1)\sin\frac{1}{N+1} + S_3$$

$$= \sin 1 - (N+1)\sin\frac{1}{N+1} + S_3$$

$$S_1 = \sum_{n=1}^{N} n\sin\frac{1}{n} = 1\cdot\sin\frac{1}{1} + 2\sin\frac{1}{2} + 3\sin\frac{1}{3} + \dots + N\sin\frac{1}{N}$$

$$S_2 = \sum_{n=1}^{N}(n+1)\sin\frac{1}{n+1} = 2\sin\frac{1}{2} + 3\sin\frac{1}{3} + \dots + N\sin\frac{1}{N} + (N+1)\sin\frac{1}{N+1}$$

$$\sum_{n=1}^{N} n\left(\sin\frac{1}{n} - \sin\frac{1}{n+1}\right) = \sin 1 - (N+1)\sin\frac{1}{N+1} + S_3$$

$$= \sin 1 - \underbrace{\lim_{N\to\infty}(N+1)\sin\frac{1}{N+1}}_{1} + \sum_{n\geq 1}\sin\frac{1}{n+1}$$

$$= \sin 1 - 1 + \sum_{n\geq 1}\sin\frac{1}{n+1} \quad \text{„like"} \quad \sum_{n\geq 1}\frac{1}{n+1} = \infty \text{ div.}$$

d) $\displaystyle\sum_{n\geq 2}\frac{\ln n^2}{n^2} = 2\cdot\sum_{n\geq 2}\frac{\ln n}{n^2}$ has the same nature as $\displaystyle\int_2^\infty \frac{\ln x}{x^2}\,dx.$

≈ 1p

$$\int_2^\infty \frac{\ln x}{x^2}\,dx = \int_2^\infty\left(-\frac{1}{x}\right)'\cdot\ln x\,dx = -\frac{1}{x}\ln x\Big|_2^\infty + \int_2^\infty \frac{1}{x}\cdot\frac{1}{x}\,dx =$$

$$= -\lim_{x\to\infty}\frac{\ln x}{x} + \frac{\ln 2}{2} + \int_2^\infty x^{-2}\,dx = 0 + \frac{\ln 2}{2} + \frac{x^{-1}}{-1}\Big|_2^\infty =$$

$$= \frac{\ln 2}{2} - \frac{1}{x}\Big|_2^\infty = \frac{\ln 2}{2} - \frac{1}{\infty} + \frac{1}{2} = \frac{\ln 2 + 1}{2} \implies \text{convergent series.}$$

$$\int_2^\infty \frac{\ln x^2}{x^2}\,dx = \int_2^\infty\left(-\frac{1}{x}\right)'\cdot\ln(x^2)\,dx = -\frac{1}{x}\ln x^2\Big|_2^\infty - \int_2^\infty \frac{1}{x}\cdot 2x\cdot\frac{1}{x^2}\,dx =$$

$$= \underbrace{\lim_{x\to\infty} \frac{-\ln x^2}{x}}_{0} + \frac{1}{2}\ln 4 - 2\int_2 \frac{1}{x^2}\,dx$$

$$= 0 + \frac{1}{2}\ln 4 - 2\cdot \frac{x^{-1}}{-1}\Big|_2^\infty$$

$$= \frac{1}{2}\ln 4 + 2\cdot\frac{1}{x}\Big|_2^\infty = \frac{1}{2}\ln 4 + 2\cdot 0 - 2\cdot\frac{1}{2} = \frac{\ln 4 - 2}{2} = \frac{\ln 2^2 - 2}{2}$$

?

$$= \frac{2\ln 2 - 2}{2} = \ln 2 - 1.$$

Method 2: Cauchy Condensation test.

Same nature as $\sum_{m\geq 2} \cancel{2^m}^{1} \cdot \frac{\ln 2^{2m}}{\cancel{2^{2m}}\, 2^m} = \sum_{m\geq 2} \frac{2m\cdot\ln 2}{2^m}$ convergent by Ratio Test.

3. $\int_0^\infty x^3\cdot e^{-x^2}dx = I$

1p

Let $u = x^2$ $\qquad x = 0 \Rightarrow u = 0$

$\qquad x = \infty \Rightarrow u = \infty$

$du = 2x\,dx \Rightarrow x\,dx = \frac{1}{2}du.$

$$I = \int_0^\infty x^2\cdot e^{-x^2}\cdot x\,dx = \int_0^\infty u\cdot e^{-u}\cdot\frac{1}{2}du = \frac{1}{2}\int_0^\infty u\cdot(-e^{-u})'\,du =$$

$$= \frac{1}{2}\cdot\left(-u\cdot e^{-u}\Big|_0^\infty + \int_0^\infty e^{-u}\,du\right)$$

$$= \frac{1}{2}\left(-u\cdot e^{-u}\Big|_0^\infty + e^{-u}\Big|_0^\infty\right)$$

$$= -\frac{1}{2}\underbrace{\lim_{u\to\infty}\frac{u}{e^u}} + 0 - \frac{1}{2}e^{-u}\Big|_0^\infty$$

$$= -\frac{1}{2}\cdot 0 + 0 - \frac{1}{2}\cdot\left(e^{-\infty} - e^0\right)$$

$$= -\frac{1}{2}e^{-\infty} + \frac{1}{2}.$$

$$= -\frac{1}{2}\cdot\frac{1}{e^\infty} + \frac{1}{2}$$

$$= 0 + \frac{1}{2}$$

$$= \frac{1}{2}$$

4. a) $\cos^2 x = \dfrac{1+\cos 2x}{2}$

$\simeq 0{,}9$

$$\cos x = \sum_{m=0}^{\infty} (-1)^m \cdot \frac{x^{2m}}{(2m)!}$$

$$\cos 2x = \sum_{m=0}^{\infty} (-1)^m \cdot \frac{(2x)^{2m}}{(2m)!}$$

$$= 1 + \sum_{m=1}^{\infty} (-1)^m \cdot \frac{2^{2m} \cdot x^{2m}}{(2m)!}$$

$$\cos^2 x = \frac{1 + 1 + \sum \cdots}{2}$$

$$= 1 + \frac{1}{2} \sum_{m=1}^{\infty} (-1)^m \cdot \frac{2^{2m} \cdot x^{2m}}{(2m)!}$$

$$= 1 + \sum_{m=1}^{\infty} (-1)^m \cdot \frac{2^{2m-1} \cdot x^{2m}}{(2m)!}$$

<u>or</u>

$f(x) = \cos^2 x$ $\qquad$ $f(0) = 1.$

$f'(x) = -2\sin x \cos x = -\sin 2x$ $\qquad$ $f'(0) = 0.$

$f''(x) = -2\cos 2x$ $\qquad$ $f''(0) = -2$

$f'''(x) = 2^2 \sin 2x$ $\qquad$ $f'''(0) = 0$

$f^{(4)}(x) = 2^3 \cdot \cos 2x$ $\qquad$ $f^{(4)}(0) = 2^3$

$f^{(5)}(x) = -2^4 \sin 2x$ $\qquad$ $f^{(5)}(0) = 0.$

$\vdots$

$f^{(2k+1)}(0) = 0.$

$f^{(2k)}(0) = (-1)^k \cdot 2^{2k-1}$

$$\sum_{m \geq 0} \frac{f^{(m)}(0)}{m!} \cdot x^m \overset{m=2k}{=\!=} \sum_{k \geq 0} \frac{f^{(2k)}(0)}{(2k)!} \cdot x^{2k} =$$

$$= \sum_{k \geq 0} \frac{(-1)^k \cdot 2^{2k-1}}{(2k)!} \cdot x^{2k} = \sum_{m \geq 1} \frac{(-1)^m \cdot 2^{2m-1}}{(2m)!} \cdot x^{2m}$$

! b) Find the higher-order derivatives $f^{(m)}(0)$, $m \in \mathbb{N}$ for :

$$f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0. \end{cases}$$

Taylor series: $f(x) = \sum\limits_{n\geq 0} f^{(n)}(0)\cdot \frac{x^n}{n!}$

To find $f^{(n)}(0)$ we can use

$$\sin x = \sum_{n\geq 0} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} \Rightarrow \frac{\sin x}{x} = \sum_{n\geq 0} (-1)^n \cdot \frac{x^{2n}}{(2n+1)!}$$

?

$$\Rightarrow f^{(2n+1)}(0) = 0$$

$$f^{(2n)}(0) = \frac{(-1)^n}{2n+1}$$

5. Radius of convergence, sum: $\sum\limits_{n\geq 1} (-1)^n \cdot n^2 \cdot x^n$

1.5p

$$L = \lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \left| \frac{\cancel{(-1)^{n+1}}^{(-1)} \cdot (n+1)^2}{\cancel{(-1)^n}_{1} \cdot n^2} \right| = \lim_n \left( \frac{(n+1)^2}{n^2} \right) = 1 = L.$$

$$R = \frac{1}{L} = 1 \Rightarrow \text{abs. conv}, (\forall)\, x \in (-1,1).$$

For $x \in (-1,1)$: $\sum\limits_{n\geq 0} (-1)^n \cdot x^n = \sum\limits_{n\geq 0} (-x)^n = \frac{1}{1-(-x)} = \frac{1}{1+x}$

Diff $\Rightarrow \sum\limits_{n\geq 1} (-1)^n \cdot n \cdot x^{n-1} = -\frac{1}{(1+x)^2} \quad \Big| \cdot x$

$$\Rightarrow \sum_{n\geq 1} (-1)^n \cdot n \cdot x^n = -\frac{x}{(1+x)^2}$$

Diff $\Rightarrow$

$$\sum_{n\geq 1} (-1)^n \cdot n^2 \cdot x^{n-1} = -\frac{(1+x)^2 - x\cdot 2(1+x)}{(1+x)^4} = -\frac{\cancel{(1+x)}^{1}(1+x-2x)}{(1+x)^{\cancel{4}\,3}} = -\frac{1-x}{(1+x)^3} \quad \Big| \cdot x$$

$$\Rightarrow \sum_{n\geq 1} (-1)^n \cdot n^2 \cdot x = \frac{x(x-1)}{(1+x)^3}.$$